

Chapter 16

Lecture 16: Uniform continuity. Lebesgue Number.

Definition 16.0.1. Two norms $\|\cdot\|_a$ and $\|\cdot\|_b$ are equivalent if there exist positive constants C_1 and C_2 such that for all vectors x ,

$$C_1\|x\|_a \leq \|x\|_b \leq C_2\|x\|_a.$$

Property 16.0.1. Equivalence of norms is an equivalence relation.

Theorem 16.0.1. All norms on $\mathbb{R}^n$ are equivalent.

Proof

Given $\|\cdot\|_a$ and $\|\cdot\|_b$ are two norms on $\mathbb{R}^n$. We define a function $f : \mathbb{R}^n \setminus \{0\} \rightarrow \mathbb{R}$ by

$$f(x) = \frac{\|x\|_b}{\|x\|_a} > 0. \quad (16.1)$$

which is continuous on $\mathbb{R}^n \setminus \{0\}$.

Since $f(x) = f(\lambda x)$ for any $\lambda > 0$, we say that $f(x)$ is completely determined by its values on the unit sphere $S = \{x \in \mathbb{R}^n : \|x\|_a = 1\}$. Note that S is compact, so $f(x)$ attains its minimum and maximum on S , say m and M . Thus, for any $x \in \mathbb{R}^n \setminus \{0\}$, we have

$$0 < m \leq f(x) \leq M < \infty, \quad (16.2)$$

which implies

$$m\|x\|_a \leq \|x\|_b \leq M\|x\|_a. \quad (16.3)$$

So the two norms are equivalent. ■

Remark In ∞ -dimensional space, the sphere is not compact. However, it is closed and bounded.

Exercise Give an example of norms on l_1 (convergent series) $l_1 = \{(x_1, x_2, \dots) \mid \sum |x_i| < \infty\}$.

Now let's come back to uniform continuity. Goal: Prove that any continuous map $f : (X, d_X) \rightarrow (Y, d_Y)$ with X compact is uniformly continuous.

Definition 16.0.2. Let X be a metric space. $A \subseteq X$. Then $\text{diam}(A) = \sup\{d(x, y) : x, y \in A\}$ is called the diameter of A .

Lemma 16.0.1 (the Lebesgue Number Lemma). Let (X, d) be a compact metric space, and let $\mathcal{A}$ be an open cover of X . Then there exists a positive number $\delta > 0$ (called a Lebesgue number for the cover $\mathcal{A}$) such that for every subset $Y \subseteq X$ with diameter less than δ (i.e., for all $x, y \in Y$, $d(x, y) < \delta$), there exists an open set $A \in \mathcal{A}$ that contains Y .

Proof of the Lemma

If $X \in \mathcal{A}$, there's nothing to prove.

Otherwise take a finite subcovering $\{A_1, A_2, \dots, A_n\}$ of $\mathcal{A}$. For each i , let $C_i = X \setminus A_i$ which is closed. We know that a closed subset of a compact set is compact, so each C_i is compact. Define

$$f(x) = \frac{1}{n} \sum_{i=1}^n d(x, C_i) \quad (16.4)$$

be the average distance from x to the closed sets C_i .

We show that $f(x) > 0$ for all $x \in X$.

For all $x \in X$, we choose A_i such that $x \in A_i$. Choose $\epsilon > 0$ such that $U_\epsilon(x) \subseteq A_i$. Then $d(x, C_i) \geq \epsilon > 0$. So $f(x) \geq \frac{1}{n}d(x, C_i) \geq \frac{\epsilon}{n}$. Let $\delta = \min_{x \in X} f(x) > 0$ (because f is continuous on compact set X). We show that δ is a Lebesgue number.

Take $B \subseteq X$ with $\text{diam}(B) < \delta$. Then we can take $x_0 \in B$ with $B \subseteq U_\delta(x_0)$. Then we can take C_m such that

$$\delta \leq f(x_0) \leq d(x_0, C_m). \quad (16.5)$$

for some m . So we have

$$U_\delta(x_0) \subseteq A_m = X \setminus C_m. \quad (16.6)$$

So $B \subseteq A_m$. ■

Example 16.0.1. Let $X \subseteq \mathbb{R}$ be covered by (a_α, b_α) . Take a finite subcovering (a_i, b_i) , $i = 1, 2, \dots, n$. Then $\delta = \min\{b_i - a_i : i = 1, 2, \dots, n\}$ is a Lebesgue number.

Definition 16.0.3. A function between two metric spaces $f : (X, d_X) \rightarrow (Y, d_Y)$ is said to be uniformly continuous if for every $\epsilon > 0$, there exists a $\delta > 0$ such that for all $x_1, x_2 \in X$, if $d_X(x_1, x_2) < \delta$, then $d_Y(f(x_1), f(x_2)) < \epsilon$.

Theorem 16.0.2. Let $f : X \rightarrow Y$ be a continuous function between two metric spaces. If X is compact, then f is uniformly continuous.

Proof

Given $\epsilon > 0$, take the open covering of Y by balls $B(y, \frac{\epsilon}{2})$ of radius $\frac{\epsilon}{2}$. Let $\mathcal{A}$ be the open covering of X by the inverse images of these balls under f . Choose δ to be a Lebesgue number for the covering $\mathcal{A}$. Then if x_1 and x_2 are any two points in X with $d_X(x_1, x_2) < \delta$, the set $\{x_1, x_2\}$ has diameter less than δ , so there exists an open set $U = f^{-1}(B(y, \frac{\epsilon}{2}))$ in $\mathcal{A}$ that contains both x_1 and x_2 . So $f(x_1)$ and $f(x_2)$ both lie in $B(y, \frac{\epsilon}{2})$, which implies that

$$d_Y(f(x_1), f(x_2)) < \epsilon. \quad (16.7)$$

Definition 16.0.4. If X is a space, a point x of X is said to be an isolated point of X if the one-point set $\{x\}$ is open in X . ■

Theorem 16.0.3. Let X be a non-empty compact Hausdorff space. If X has no isolated points, then X is uncountable.

Proof

Step 1 We show first that given any non-empty open set U of X and any point x of X , there exists a non-empty open set V contained in U such that $x \notin \bar{V}$.

Choose a point y of U different from x ; this is possible if x is in U because x is not an isolated point of X and it is possible if x is not in U simply because U is non-empty. Since X is Hausdorff, there exist disjoint open sets W_x and W_y containing x and y , respectively. Then the open set $V = U \cap W_y$ is non-empty, is contained in U , and its closure $\bar{V}$ is contained in the complement of W_x , so $x \notin \bar{V}$.

Step 2 We show that given $f : \mathbb{Z}_+ \rightarrow X$, the function f is not surjective. It follows that X is uncountable.

Let $x_n = f(n)$. Apply Step 1 to the non-empty open set X to choose a non-empty open set $V_1 \subseteq X$ such that $x_1 \notin \bar{V}_1$. In general, given V_{n-1} open and non-empty, choose V_n to be a non-empty open set such that $V_n \subseteq V_{n-1}$ and $x_n \notin \bar{V}_n$. Then we have a sequence of non-empty closed sets $\bar{V}_n$ with

$$\bar{V}_1 \supseteq \bar{V}_2 \supseteq \bar{V}_3 \supseteq \dots \quad (16.8)$$

Since X is compact, any collection of closed subsets of X with the finite intersection property has non-empty intersection. Thus, there exists a point y in the intersection of all the $\bar{V}_n$. By construction, $y \neq x_n$ for all n , so f is not surjective. ■

Corollary 16.0.1. $[a, b] \subseteq \mathbb{R}$ is uncountable.